

Skin Analysis Feature — Complete Technical Specification

****Version:** 3.0**

****Date:** 2026-04-28**

****Owner:** RARE reflection**

****Status:** Production-Ready**

1. Overview

The Skin Analysis feature is a free, AI-powered tool that analyses a user's skin via a photo and a 25-question questionnaire, then provides personalised product recommendations from our catalog. It serves as a customer acquisition hook and drives revenue through brand partnerships.

****Core flow (happy path):****

1. User uploads a selfie → Open-source ML model (Arko007) detects cosmetic skin conditions.
2. User completes a 25-question questionnaire covering oiliness, acne, pigmentation, sensitivity, habits, and budget.
3. Backend pre-filters product catalog by budget, trigger ingredients, and fungal acne safety.
4. Pre-processed user profile + filtered catalog → conditionally sent to DeepSeek API based on profile changes.
5. If a significant change is detected:
 - DeepSeek generates prioritised concerns and recommended ingredients
 - Groq LLaMA 3.1 8B ranks and personalises product recommendations
6. If no significant change:
 - Recommendations are generated directly from stored profile data (no LLM call)
7. First analysis stores the user's skin profile; subsequent visits generate instant product recommendations at zero marginal cost unless re-analysis is triggered.

If a severe clinical concern is detected by the ML, the flow is ****ethically triaged**** to a health-advisory screen, skipping cosmetic recommendations entirely and offering a dermatologist connection CTA.

2. Architecture & Data Flow

[User] —(photo)—> [ML API] → conditions_detected

—(questionnaire)—> Frontend → User Profile JSON

—> Backend Pre-filtering → Curated Catalog

—> Conditional Engine:

IF significant_change:

→ DeepSeek API → ingredients + concerns

→ Groq LLaMA → ranking

ELSE:

→ DB retrieval (no LLM)

—> Post-processing & Storage

—> "Many Suggestions" Engine

****Key design principles:****

- Significant change is defined as:

- Change in primary concern OR
- 6 or more questionnaire answers differ from stored profile

- LLM is called ****only on first analysis**** or when questionnaire answers change significantly.
- Unsafe products ****never**** reach the LLM; hard exclusions happen in backend pre-filtering.
- Ingredients first, then products: the LLM derives actives, then matches catalog products that contain those actives.
- Ethical triage: severe clinical ML detections reroute the entire experience to a health-first advisory.

3. Step-by-Step Implementation Details

3.1 ML Image Analysis

- **Model:** Arko007/skin-disease-detector-ai (Hugging Face)
- **Deployment:** FastAPI endpoint on Render/Railway (or your preferred platform)
- **Input:** JPEG/PNG image bytes
- **Output:**

```
```json
{
 "conditions_detected": ["acne", "hyperpigmentation", "enlarged_pores"],
 "confidence_scores": {
 "acne": 0.87,
 "hyperpigmentation": 0.72,
 "enlarged_pores": 0.65
 }
}
```
```

3.1.1 Ethical Triage for Clinical Detections

Arko007 can detect severe clinical conditions (e.g., melanoma, basal cell carcinoma). We must **never** pass these labels to the LLM for cosmetic recommendations. Instead, we implement an ethical triage that builds trust and product-market-fit (PMF) data.

Severe labels (configurable):

```
```python
SEVERE_LABELS = [
 "melanoma", "basal_cell_carcinoma", "squamous_cell_carcinoma",
 "psoriasis_severe", "actinic_keratosis"
]
```
```

Backend logic (`app.py`):

```
```python
def
has_severe_condition(ml_result):
 return any(cond.lower() in SEVERE_LABELS
 for cond in ml_result.get("conditions_detected", []))

if has_severe_condition(ml_result):
 # Skip all skincare analysis.
 # Store the detection for analytics.
 return {
 "status": "clinical_concern_detected",
 "message": (
 "Our scan detected an irregular texture that falls outside the "
```

```

 "scope of cosmetic skincare. We always prioritize your health over "
 "retail. Would you like us to connect you with a board-certified "
 "dermatologist?"
),
 "dermatologist_cta": True
}
...

```

**\*\*Frontend experience:\*\***

When receiving `status: "clinical\_concern\_detected"`, render a dedicated **\*\*Health Advisory Screen\*\***:

- **\*\*Headline:\*\*** “Your skin’s health comes first.”
- **\*\*Body:\*\*** The message above, in calm, warm typography.
- **\*\*Primary CTA button:\*\*** “Find a Dermatologist” (or “Connect Me with a Specialist”)
- **\*\*Secondary link:\*\*** “Continue to cosmetic recommendations” (allows users to proceed anyway; tracked separately)

**\*\*Design:\*\*** Matte, honey/amber tones. No red or alarming visuals. A soft illustration of a dermoscope or a calm human face. The brand voice is reassuring, not fearful.

**\*\*PMF Tracking via Button CTR:\*\***

Emit analytics events:

- `health\_advisory\_shown` – fired when the advisory screen is displayed.
- `dermatologist\_cta\_click` – primary CTA.
- `continue\_to\_cosmetic\_click` – secondary link.

**\*\*Data interpretation:\*\***

- **\*\*>20% click-through on “Find a Dermatologist”:\*\*** Strong PMF signal for a clinical telehealth integration. You can build or partner with a dermatologist booking service.
- **\*\*<5% click-through:\*\*** Most users want the cosmetic experience; the advisory serves as a trust builder.
- **\*\*Secondary clicks:\*\*** Indicate users who want the cosmetic routine despite the advisory—use to refine messaging.

This transforms a liability into a strategic PMF sensor, zero legal risk, and positions the brand as ethical.

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### ### 3.2 Questionnaire UI Specification

The questionnaire collects 25 answers across 6 groups, presented in **\*\*4 seamless screens\*\*** to reduce fatigue.

### #### 3.2.1 UX Design & Screen Breakdown

Screen	Title	Questions	Visual Theme
1	<b>**Your Skin Rhythm**</b>	Q4–Q8 (oiliness, hydration, water intake)	Soft watercolor textures, warm terracotta
2	<b>**Your Glow &amp; Flow**</b>	Q1–Q3 (profile), Q13–Q16 (pigmentation, texture, sun)	Golden-hour gradient, subtle shimmer
3	<b>**Skin Language**</b>	Q9–Q12 (acne), Q17–Q20 (sensitivity)	Calm amber, minimal
4	<b>**Your Routine, Your Way**</b>	Q21–Q25 (habits, budget, concern, decision driver)	Clean slate, gentle highlights

#### **\*\*Design principles:\*\***

- Background: cream/off-white (never pure white).
- Palette: terracotta (#C67B5C), honey (#D4A373), amber (#B08968), soft sage (#A3B18A).
- Progress indicator: subtle step dots (no percentage, no timer).
- Microcopy: “A few details so we can personalise your results — this won’t take long.”
- No asterisks or “required” labels; all questions optional except Q24 (primary concern).

### #### 3.2.2 Full Question List

(Include all 25 questions exactly as previously described, with the pill options and special notes like color swatches for Q3, highlight for Q12, note for Q18, free text for Q19. I'll condense but maintain completeness.)

#### **\*\*Group 1 — Basic Profile\*\***

- Q1: Gender (Single select: Female, Male, Prefer not to say)
- Q2: Age range (Under 18, 18–24, 25–34, 35–44, 45+)
- Q3: Skin tone with colour swatches (Fair/Light, Medium/Wheatish, Olive/Tan, Deep/Dark)

#### **\*\*Group 2 — Skin Type: Oiliness & Hydration\*\***

- Q4: Post-wash feel (Tight and dry, Comfortable and balanced, Dry on cheeks normal elsewhere, Gets oily quickly all over, Oily on forehead/nose dry on cheeks)
- Q5: T-zone shine speed (Within 30 mins, After 1–2 hours, By end of day, Rarely or never)
- Q6: AC room reaction (Gets flaky/very dry, Feels tight but okay, No change, Gets itchy/irritated)
- Q7: Moisturiser tightness after 30 min (Yes always, Sometimes, No feels fine, I don't use moisturiser)
- Q8: Daily water intake (<4 glasses, 4–6 glasses, 6–8 glasses, 8+ glasses)

#### **\*\*Group 3 — Acne & Blemishes\*\***

- Q9: Pimple frequency (Almost never, Once a month or less, Few times a month, Every week, Almost always have some)
- Q10: Breakout triggers (Multi-select: Stress/poor sleep, Oily/junk food, Before my period, After new products, Random, I rarely break out)

- Q11: Dark mark persistence (Yes stays months, Yes fade in weeks, Rarely, No fade quickly)
- Q12: Acne appearance (Multi-select: Blackheads, Whiteheads, Red inflamed, Deep cysts, Tiny flesh-coloured bumps **\*\*highlighted\*\***, I don't have acne). Show small visual examples; tiny bumps trigger fungal acne filter.

**\*\*Group 4 — Pigmentation, Tone & Texture\*\***

- Q13: Dark spots (Yes very noticeable, A few spots, Very faint, No tone even) - Q14: Dullness (Yes almost always, Sometimes, Rarely, No skin bright)
- Q15: Rough texture (Yes rough and bumpy, Slightly uneven, Mostly smooth, Very smooth)
- Q16: Sun exposure (Mostly indoors, ≤30 mins, 1–2 hours, 3+ hours outdoors)

**\*\*Group 5 — Sensitivity & Reactions\*\***

- Q17: Reaction to new products (Yes very easily, Sometimes, Rarely, Never)
- Q18: Diagnosed conditions (Multi-select: Eczema, Rosacea, Psoriasis, Seb. dermatitis, Melasma, None). If any selected, show disclaimer note immediately. - Q19: Known triggers (Multi-select: Fragrance/Parfum, Alcohol, Retinol, AHAs/ BHAs, Essential oils, Sulphates, None). Free-text “+ Add another ingredient...” below.
- Q20: Switching reaction (Fine adapts quickly, Needs few days, Often breaks out, Rarely switches)

**\*\*Group 6 — Habits & Preferences\*\***

- Q21: Skincare frequency (Morning & night, Once a day, Few times a week, Rarely, Just starting out)
- Q22: Sunscreen usage (Every day, Only when outside, Occasionally, Never)
- Q23: Monthly budget (Under ₹500, ₹500–1,500, ₹1,500–3,000, ₹3,000+)
- Q24: Primary concern (Acne & breakouts, Dark spots & pigmentation, Dryness & dehydration, Oiliness & shine, Dullness – more glow, Anti-ageing & fine lines, Sensitive/reactive skin) **\*\*Highlighted\*\***
- Q25: Decision driver (Ingredient list, Customer reviews, Dermatologist recommended, Brand name, Price/value)

**#### 3.2.3 Questionnaire → JSON Mapping**

(Full mapping table as previously provided. I'll include it here for completeness, referencing all internal value strings. Since the document is long, I'll keep it succinct but explicit.)

JSON field	Q#	Mapping (pill → value)
`profile.gender`	Q1	Female→`female`, Male→`male`, Prefer not to say→`undisclosed`
`profile.age_range`	Q2	e.g., 25–34→`25-34`
`profile.skin_tone`	Q3	Medium/Wheatish→`medium_wheatish`, Deep/Dark→`deep_dark`, etc.

`skin\_type.\*`	Q4-8	e.g., Oily on forehead/nose, dry on cheeks→`oily\_tzone\_dry\_cheeks`	
`acne.frequency`	Q9	Few times a month→`few\_times\_a\_month`, etc.	
`acne.triggers[]`	Q10	Stress or poor sleep→`stress`, Before my period→`hormonal`, etc.	
`acne.leaves\_dark\_marks`	Q11	Yes, stays for months→`yes\_stays\_months`	
`acne.acne\_type[]`	Q12	Tiny flesh-coloured bumps→`tiny\_flesh\_coloured\_bumps` (triggers fungal filter)	
`pigmentation\_texture.\*`	Q13-16	e.g., A few spots→`a\_few\_spots`	
`sensitivity.\*`	Q17-20	e.g., Fragrance/Parfum→`fragrance`, Eczema→`eczema`	
`habits.\*`	Q21-25	e.g., ₹500–1,500→`500\_to\_1500`, Dark spots & pigmentation→`dark\_spots\_pigmentation`	

**\*\*Note:\*\*** Multi-select fields become arrays. Q19 custom entries are appended as lower-case strings. The frontend must maintain a shared `q\_mapping.json` config to ensure consistency.

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### ### 3.3 Backend Pre-Filtering (Before LLM Call)

**\*\*Objective:\*\*** Reduce token cost and enforce hard safety rules.

```

python def pre_filter_catalog(user_profile,
all_products):
 # 1. Budget filter max_price =
 BUDGET_LIMITS[user_profile["habits"]["monthly_budget_inr"]] products = [p for p in
all_products if p["price_inr"] <= max_price]

 # 2. Hard-exclude trigger ingredients (with synonym mapping) triggers =
 expand_synonyms(user_profile["sensitivity"]["known_triggers"]) products = [p for p in
products
 if not any(ing in triggers for ing in p["key_ingredients"])]

 # 3. Fungal acne filter (if tiny_flesh_coloured_bumps in acne_type) if
"tiny_flesh_coloured_bumps" in user_profile["acne"]["acne_type"]:
 fungal_blacklist = {"coconut_oil", "argan_oil", "olive_oil", "fatty_acids",
"squalane", "isopropyl_myristate", "oleic_acid", "ester"} products = [p for p in
products
 if not any(ing in fungal_blacklist for ing in p["key_ingredients"])]

 # 4. Limit to top 30-50 products by overlap with ML-detected concerns
 concerns = user_profile["ml_detections"]["conditions_detected"]

```

```
products.sort(key=lambda p: -sum(1 for c in concerns if c in
p["concerns_addressed"])) return products[:50] ```
```

**\*\*Expanded synonym mapping:\*\***

```
```python
TRIGGER_SYNONYMS = {
    "fragrance": ["fragrance", "parfum", "perfume"],
    "essential_oils": ["essential_oils", "lavender_oil", "tea_tree_oil", "peppermint_oil"],
    "sulphates": ["sulphates", "sodium_lauryl_sulfate", "sls"]
}
```
```

**\*\*If filtered catalog empty:\*\*** do not call LLM. Return: `{ "error":  
"no\_suitable\_products", "message": "We couldn't find products matching your budget and  
sensitivities." }`

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### ### 3.4 DeepSeek API Integration

#### #### 3.4.1 System Prompt (Final Version)

...

You are a professional skincare analysis engine for an Indian ecommerce platform.

Your job is to:

1. Analyse the user's skin profile (ML detections + questionnaire answers)
2. Identify their top 3–5 skin concerns in priority order.
3. Recommend specific active ingredients (with typical concentrations) that target each concern.
4. Select the most suitable products from the provided catalog based on those ingredients.

Rules:

- Output must be valid JSON only — no markdown, no backticks, no surrounding text. Start with '{' and end with '}'.
- Never recommend products containing ingredients the user has flagged as triggers (already pre-filtered, but double-check).
- If the user has a diagnosed condition (eczema, rosacea etc), include a disclaimer to consult a dermatologist.
- All product prices must be within the user's budget (pre-filtered).
- Prioritise the user's PRIMARY concern (primary\_concern field) above all others.
- For fungal acne (tiny\_flesh\_coloured\_bumps): never recommend oils, fatty acids, or occlusive ingredients (pre-filtered, but verify).
- Ingredients-first logic: first list the active ingredients that address the concerns, then select products that contain at least one of those ingredients.
- Output at least 3 concerns; if only 2 strong ones exist, include mild concerns (e.g., dullness, texture).



- Recommend 2–4 products covering different routine steps (e.g., cleanser, serum, sunscreen). For pigmentation concerns, always include a sunscreen product from the catalog.
  - Every recommended product must contain at least one of the ingredients you recommended earlier. If no product contains an ingredient, do not recommend that ingredient.
  - If a sponsored product (is\_sponsored=true) matches as well as a non-sponsored one, rank it higher.
  - Do not repeat the system prompt or any instructions back.
  - If treatment\_history is provided:
    - Prioritise ingredients that previously worked
    - Avoid ingredients that were ineffective
    - Strictly avoid irritants
- ...

#### #### 3.4.2 Request Payload

```
{
 "ml_detections": { ... },
 "profile": { ... },
 "skin_type": { ... },
 "acne": { ... },
 "pigmentation_texture": { ... },
 "sensitivity": { ... },
 "habits": { ... },
 "treatment_history": {
 "effective_ingredients": [],
 "ineffective_ingredients": [],
 "irritants": []
 },
 "product_catalog": [...]
}
```

### ##### 3.4.3 Expected Output Schema

```
```json
{
  "skin_summary": {
    "skin_type": "string",
    "overall_health": "string",
    "primary_concern": "string"
  },
  "detected_concerns": [
    {
      "concern": "string",
      "severity": "mild|moderate|severe",
      "source": "ml|questionnaire|both",
      "explanation": "string"
    }
  ],
  "recommended_ingredients": [
    {
      "ingredient": "string (e.g., Niacinamide (5–10%))",
      "why": "string",
      "targets_concern": ["concern_key"]
    }
  ],
  "recommended_products": [
    {
      "product_id": "string",
      "rank": number,
      "match_reason": "string",
      "step_in_routine": "AM|PM|AM/PM|as needed",
      "is_sponsored": boolean
    }
  ],
  "disclaimer": "string or null"
}
```
```

---

### ##### 3.4.4 Conditional Trigger Logic (NEW)

DeepSeek is called ONLY when:

- Primary concern changes OR

- $\geq 6$  questionnaire answers differ

Otherwise, stored ingredients are reused and no LLM call is made.

### ### 3.5 Groq LLaMA 3.1 8B Ranking Layer (NEW)

Purpose:

- Rank and personalise product recommendations

Trigger:

- First analysis
- Significant profile change

Input:

- User profile
- Derived ingredients
- Candidate products
- User feedback

Output:

- Ranked product list with reasoning

Note:

- Not used for returning users without significant change

### ### 3.6 Post-Processing & Validation

After receiving the LLM response, validate it before storing/displaying:

```
```python def validate_llm_output(output, user_payload,
sent_catalog):
    # 1. JSON is valid (already parsed) # 2. Required fields present    assert all(k
in output for k in ["skin_summary", "detected_concerns",
                    "recommended_ingredients",
"recommended_products"])
    # 3. Min 2 concerns, min 2 products (warn if less than 2 products)    assert
len(output["detected_concerns"]) >= 2    if
len(output["recommended_products"]) < 2:        log_warning("Only 1 product
recommended")

    # 4. All products exist in catalog and match an ingredient
rec_ingredients_lower = [extract_ingredient_name(i["ingredient"])
for i in output["recommended_ingredients"]]    for rp in
output["recommended_products"]:
        prod = find_product(rp["product_id"], sent_catalog)        if not
prod:
            raise ValueError(f"Product {rp['product_id']} not in catalog")        if not any(ing in
prod["key_ingredients"] for ing in rec_ingredients_lower):
            raise ValueError(f"Product {rp['product_id']} doesn't contain recommended
ingredient")

    # 5. Pigmentation primary concern → must include sunscreen    if
user_payload["habits"]["primary_concern"] == "dark_spots_pigmentation":
has_sunscreen = any(        find_product(p["product_id"],
sent_catalog)["category"] ==
"sunscreen"        for p in
output["recommended_products"]
    )
    if not has_sunscreen:        log_error("Missing sunscreen for
pigmentation user")

    # 6. Diagnosed conditions → disclaimer not null    if
user_payload["sensitivity"]["diagnosed_conditions"]:
        assert output["disclaimer"] is not None and len(output["disclaimer"]) > 10 ```
```

If validation fails, retry once; if still failing, fallback to a simple recommendation based on stored ingredients.

3.7 User Profile Storage

Once validated, persist the profile for subsequent visits.

****Database table:**** `skin\_profiles`

```
```sql
CREATE TABLE skin_profiles (user_id
UUID PRIMARY KEY,
 profile_data JSONB NOT NULL, -- full input + output
ingredient_list TEXT[] GENERATED ALWAYS AS (ARRAY(
 SELECT DISTINCT LOWER(LEFT(elem->>'ingredient',
 strpos(elem->>'ingredient', ' ') - 1
)
)
 FROM jsonb_array_elements(profile_data->'recommended_ingredients') elem
)
) STORED,
 created_at TIMESTAMPTZ DEFAULT NOW(), updated_at
TIMESTAMPTZ DEFAULT NOW()
);
```
```

****Profile data includes:****

- Original questionnaire answers + ML detections
- LLM output: skin_summary, detected_concerns, recommended_ingredients, initial recommended_products - Purchase history (appended later)
- Behaviour flags (favourites, cart abandons)

****Consent:**** explicit opt-in before analysis. "Delete my data" endpoint required.

3.8 "Many Suggestions" Engine

With stored ingredients and skin profile, generate multiple product carousels ****without calling the LLM****.

3.8.1 Primary Routine (Curated)

Show the 2–4 products from the last LLM analysis, with match reasons.

3.8.2 Complete Your Routine

Query: products matching stored ingredients but from categories missing in the primary routine.

```
```sql
SELECT * FROM products
WHERE category NOT IN ('serum', 'sunscreen')
AND key_ingredients && (SELECT ingredient_list FROM skin_profiles WHERE user_id = ?)
AND price_inr <= (SELECT (profile_data->'habits'->>'monthly_budget_inr')...) LIMIT 5;
```

...

#### #### 3.8.3 Alternative Products for Each Ingredient

For each ingredient (e.g., "Niacinamide"), show 3 products with the same ingredient but different price/brand.

#### #### 3.8.4 Similar Skin Cluster Recommendations

Batch-process anonymous profiles to build clusters (e.g., combination + pigmentation + budget range). Then recommend top-selling products within the same cluster.

#### #### 3.8.5 Ingredient Education Links

Link to static ingredient pages (SEO + authority).

All carousels load asynchronously on the user dashboard.

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### ### 3.8 Latency & Loading Experience (Optimistic UI)

DeepSeek API calls may take 5–15 seconds. We must not show a blank spinner.

**\*\*Solution:\*\*** After questionnaire submission, immediately redirect to a full-screen "Analysis in Progress" page.

**\*\*Phase 1 – Instant transition:\*\***

- Show the uploaded photo (blurred overlay).
- Headline: "We're building your personalised skin profile..."
- Subtext: "Our AI is cross-referencing your answers with thousands of ingredient studies."

**\*\*Phase 2 – Dynamic progress (simulated or real via SSE):\*\***

Cycle through these lines, each fading in with a checkmark:

1. "Mapping your skin type..."
2. "Filtering out ingredients that don't suit you..."
3. "Finding the best actives for your concerns..."
4. "Matching products from our clean catalog..."
5. "Finalising your personalised routine..."

Each line appears after ~1.5s. If the API takes >12s, show:

"This is taking a little longer than expected. Sit back — we're still working on your best matches."

**\*\*Phase 3 – Seamless reveal:\*\*** As soon as data arrives, slide into the results dashboard—no "loading complete" message.

**\*\*Backend support:\*\*** Optionally provide a `/status` endpoint or SSE to report progress; or simply use CSS animation timers to simulate the steps while the API call runs.

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### ### 3.9 Privacy & Compliance

- \*\*Photo is not stored.\*\* Only ML detection results and questionnaire are kept.
- Explicit consent before storage.
- Data encrypted at rest and in transit.
- Logs must not contain raw images or full LLM responses with PII.
- Right to deletion: full profile hard-delete via API.

### ### 3.10 Feedback System (NEW)

Users provide feedback:

- Worked well
- No change
- Made worse
- Caused irritation

Feedback is converted into:

- effective\_ingredients
- ineffective\_ingredients
- irritants

This is used in future DeepSeek calls to improve recommendations.

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### ### 3.11 15-Day Reanalysis Lifecycle (NEW)

- Trigger: 15 days after last analysis
- User receives email reminder
- Feedback collected

- If significant change:

→ DeepSeek + Groq triggered

- Else:

→ DB-based recommendations

## ## 4. Cost Breakdown

**\*\*Per new analysis (DeepSeek):\*\***

- System tokens: ~800
- Input tokens: ~1,200 (with 30–50 products)
- Output tokens: ~600
- **\*\*Total:\*\*** ~2,600 tokens → approx ₹0.20 per request

**\*\*Monthly (moderate traffic: 10,000 new users):\*\***

- ML hosting (Render/Railway): ₹500–1,500
- DeepSeek API: ₹2,000–2,500
- Database & backend: ₹1,000
- **\*\*Total:\*\*** ~₹4,000–5,000/month

Returning users cost ₹0 — all recommendations generated from stored profile.

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## ## 5. Implementation Timeline

Task	Est. Time
-----	-----
ML model hosting + sanitisation	1 day
Questionnaire UI + mapping	2 days
Pre-filtering pipeline	1 day
DeepSeek integration + prompt	1 day
Post-processing validation	1 day
User profile DB + storage	1 day
"Many Suggestions" endpoints	2 days
Optimistic loading UI	1 day
Health advisory screen + PMF events	1 day
QA & validation harness	2 days
<b>**Total**</b>	<b>**~12 days**</b>

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## ## 6. Validation Harness

Use the provided `deepseek\_validation\_harness.html` with 5 test profiles:



1. **Priya** – typical Indian user, combination skin, pigmentation
2. **Rahul** – fungal acne (must pass safety filter)
3. **Meena** – eczema + fragrance allergy, must show disclaimer
4. **Ananya** – deep skin tone, severe pigmentation, must include sunscreen
5. **Vikram** – beginner, under ₹500 budget

Automated checks:

- Valid JSON
- $\geq 2$  concerns,  $\geq 2$  products
- All products within budget
- No trigger ingredients appear
- Fungal acne profile: no blacklisted ingredients
- Eczema profile: disclaimer present
- Pigmentation profile: sunscreen included

Additionally, test the clinical triage with a mock Arko007 response containing "melanoma" to verify the health advisory screen appears.

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## ## 7. Analytics & PMF Tracking

All events sent to your analytics platform (e.g., Mixpanel, Amplitude, custom). | Event |

Fired when | |-----|-----|

| `analysis\_started` | User begins questionnaire |

| `questionnaire\_completed` | All answers submitted |

| `ml\_analysis\_complete` | ML model returns result |

| `clinical\_concern\_detected` | Severe condition found (health advisory shown)

|

| `dermatologist\_cta\_click` | User clicks primary CTA on advisory |

| `continue\_to\_cosmetic\_click` | User skips advisory, proceeds to cosmetics | |

| `llm\_call\_started` / `llm\_call\_success` / `llm\_call\_failed` | Track API latency & errors |

| `recommendation\_viewed` | User sees product recommendations |

| `product\_clicked` | User clicks a recommended product |

| `purchase\_from\_recommendation` | Post-click purchase |

Use `dermatologist\_cta\_click` divided by `clinical\_concern\_detected` as the key PMF metric for telehealth expansion.

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## ## 8. Sample Input & Output

(Two complete examples: Priya (normal) and the clinical triage response.)

**Example 1 – Priya (happy path)**

\_Input:\_ (as shown in previous Payload sample)

\_Output:\_ (the improved output with 3 products, sunscreen, etc.)

**\*\*Example 2 – Clinical triage (severe condition)\*\***

\_ML Output:\_ `{"conditions\_detected": ["melanoma", "acne"],

"confidence\_scores": {"melanoma": 0.68, "acne": 0.81}}`

\_Backend response:\_

```json

{

 "status": "clinical_concern_detected",

 "message": "Our scan detected an irregular texture that falls outside the scope of cosmetic skincare. We always prioritize your health over retail. Would you like us to connect you with a board-certified dermatologist?",

 "dermatologist_cta": true

}

```

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**\*End of document.\***